

COURSE TITLE : MOULDS AND DIES TECHNOLOGY
COURSE CODE : 4112
COURSE CATEGORY : B
PERIODS/WEEK : 5
SEMESTER : 4
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Plastics, Processing, Injection Moulding, Hand Injection Mould, Feed system, Ejection, Injection moulds	20
2	Compression Moulding, Transfer moulding	20
3	Other Moulding Process	20
4	Die Casting	15
	Total	75

COURSE OUTCOMES

After completion of course the student should be able to:

1. Understand the different types of plastics and processing.
2. Know the principles of injection moulding.
3. Understand the processing techniques of Thermosets.
4. Comprehending other moulding process like extrusion moulding, blow moulding, rotational moulding.
5. Understand diecast process.

SPECIFIC OUTCOMES

MODULE-I

1.1.0 Understand different types of plastics and processing

- 1.1.1 state Testing of plastics
- 1.1.2 Distinguish the plastics used in moulding industry
- 1.1.3 State the applications of plastics. types of additives.
- 1.1.4 State the purpose of moulding.

1.2.0 Know the principle of injection moulding

- 1.2.1 State the application of Injection moulding
- 1.2.2 Explain the working of Injection moulding machines
- 1.2.3 Describe the moulding calculations
- 1.2.4 Solve practical problems of Injection moulding machine
- 1.2.5 Shot weight calculation
- 1.2.6 Plasticizing calculation
- 1.2.7 Minimum cycle time calculation
- 1.2.8 Clamping force calculation
- 1.2.9 No. of cavity calculation
- 1.2.10 Clear the Basic Terminology of Hand Injection moulds
- 1.2.11 Select the mould material and their specification

1.3.0 Apply the feed system and injection moulds

- 1.3.1 Distinguish between Runner and gate
- 1.3.2 List out the types of runners
- 1.3.3 Calculate the runner diameter
- 1.3.4 List out the types of gates
- 1.3.5 Calculate the gate dimensions
- 1.3.6 Identify Ejection Techniques
- 1.3.7 Calculate the Ejection force required
- 1.3.8 Determine appropriate parting surface for a mould
- 1.3.9 Highlight the importance of venting
- 1.3.10 Illustrate the Shrinkage of plastics
- 1.3.11 Outline importance and need for controlling temperature of Injection mould
- 1.3.12 Highlight the moulds with side core and side cavity
- 1.3.13 Justify the moulds with Internal undercuts
- 1.3.14 Explain two plate and three plate mould
- 1.3.15 Explain Runner less moulds
- 1.3.16 Describe multi daylight moulds
- 1.3.17 Find the importance of mould maintenance

MODULE – II

2.1.0 Analyze the processing techniques of Thermosets

- 2.1.1 Explain the principle of compression moulding
- 2.1.2 Explain the process of compression moulding
- 2.1.3 State the applications

- 2.1.4. Describe the principle of Transfer moulding
- 2.1.5 Explain the process of Transfer moulding
- 2.1.6 List the applications, advantages and limitations of Transfer moulding

MODULE- III

3.1.0 Comprehend other moulding processes

- 3.1.1 Explain the principle of Extrusion moulding
- 3.1.2 Explain the Process of Extrusion moulding
- 3.1.3 List the applications of Extrusion moulding
- 3.1.4 Explain the Principle of Blow moulding
- 3.1.5 Explain the process of Blow moulding
- 3.1.6 State the applications of Blow moulding
- 3.1.7 Explain the principle of Rotational moulding
- 3.1.8 State the applications of Rotational moulding
- 3.1.9 Explain the Joining of Plastics

MODULE- IV

4.1.0 Understand Die casting process

- 4.1.1 State the advantages of die casting techniques
- 4.1.2 List the principle advantages and limitations of pressure die casting
- 4.1.3 Illustrate the die casting terminology
- 4.1.4 List out the different die casting alloys
- 4.1.5 State the advantages and disadvantages of different die casting alloys
- 4.1.6 Describe the practical applications
- 4.1.7 Explain the working of Die casting machines
- 4.1.8 Compare Hot Chamber and Cold Chamber Process
- 4.1.9 List the types of Die Casting dies
- 4.1.10 State the advantages and disadvantages of different types of die

CONTENT OUTCOMES

MODULE-I

Plastics in the mould industry – properties of plastics – application – processing – Injection moulding – Principle – application – Temperature range – Injection moulding machines – specifications – shot weight calculation – plasticizing capacity – clamping force calculation – minimum cycle time calculation – No. of cavity calculation – Testing of plastics-visual test, smoke test, carbon black test, metal flow index test- Hand Injection moulds – Terminology - mould materials Feed system – Runner – types of runner – Runner die calculation – Gate – types of gates – Gate calculation – Ejection Techniques – Ejection force calculation – Parting surface – venting – shrinkage – values of shrinkage –Factors governing shrinkage–

temperature control – cooling – mould cooling calculation – side core and side cavity – Determine core and cavity

dimensions-Internal undercuts – Two plate and three plate moulds – runner less moulds – multi daylight moulds – mould maintenance.

MODULE-II

Compression moulding – application – temperature and pressure ranges – basic types of compression moulds – flash type mould – positive mould – landed positive mould - semi positive mould. Transfer moulding – application – advantages and limitations – types of transfer mould – pot type and plunger type

MODULE-III

Extrusion moulding – description of the process – application. Blow moulding – Description of the process – application. Rotational moulding – description of the process – application. Casting and moulding – principle – advantages – application – calendaring – vacuum forming – perform moulding mat and fabric moulding – premix moulding – filament winding – multicolor moulding

MODULE-IV

Die casting, - advantages of die casting techniques – principle advantages and limitations of pressure die casting process – die casting terminology – die casting alloys – zinc, aluminium, magnesium, copper etc – composition – properties of different die casting alloys – die casting machines – hot chamber and cold chamber – comparison of hot chamber and cold chamber process – die casting dies.

REFERENCE

1. Introduction to Injection Moulding - Pye
2. A Text Book of Production Engineering - P.C. Sharma (SChand)
3. Manufacturing Engineering and Technology - Serope Kalpakjian
4. Fundamentals of plastics mould design-Sanjay K. Nayak-Tata Mcgrohill.